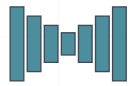


Weight perturbation across three AI weather models

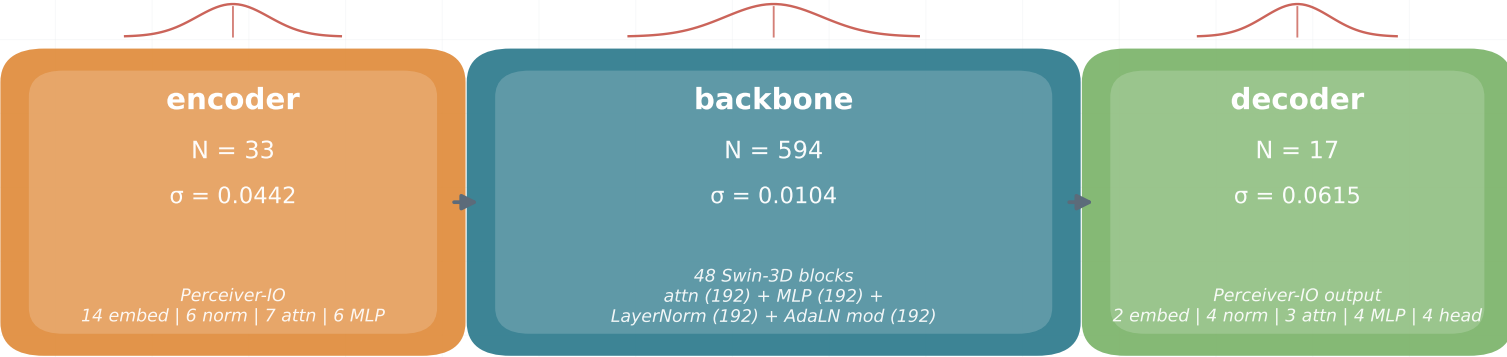
Phase 2 layer-group ablation at $\sigma_{full} = 0.01$ with $\sqrt{N_{total}/N_{group}}$ variance scaling. Multiplicative noise: $w' = w (1 + \sigma \cdot N(0,1))$.



Aurora

Swin-Transformer 3D U-Net

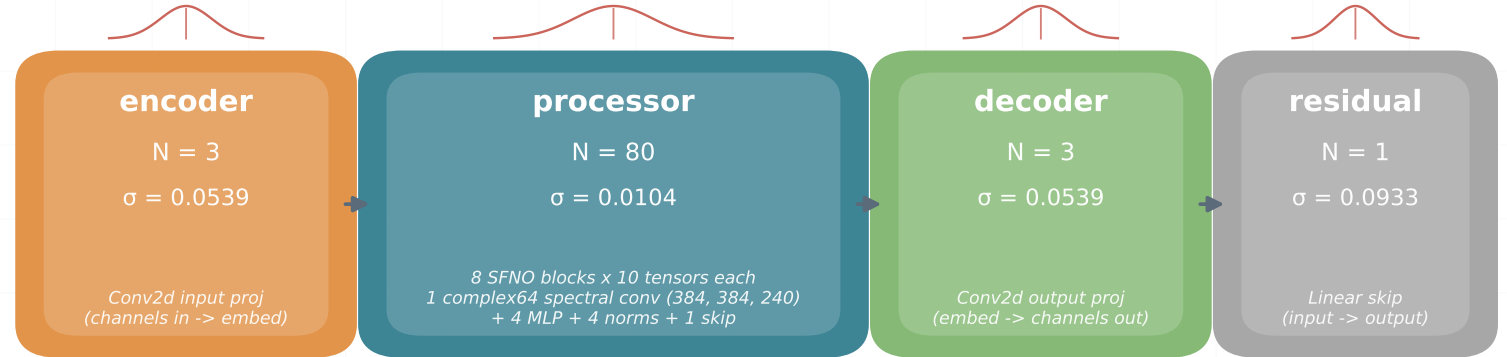
$N_{total} = 644$



SFNO

Spherical Fourier Neural Operator

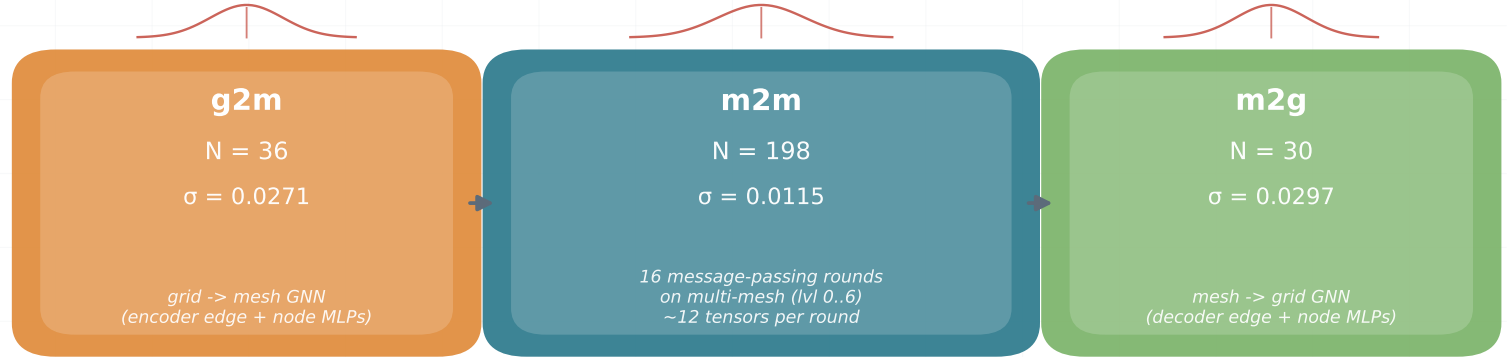
$N_{total} = 87$



GraphCast

Multi-Mesh Graph Neural Network

$N_{total} = 264$



encoder **processor** **decoder** **aux / residual** = Gaussian noise injection on every learnable tensor in the group